



Seh Na Mellick

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All models are wrong;
several of them were mine

EDUCATION

Carnegie Mellon University

Pittsburgh, PA, 2024 – Present

Ph.D. Computational Biology, (expected 2029)

- Did transposon mutagenesis screen using *Klebsiella pneumoniae* laboratory parent strain that (unbeknownst to me) had a key quorum sensing regulator knocked out. I was studying collective behaviors. Unsurprisingly, the mutants did not behave collectively.

University of Nebraska Omaha

Omaha, NE, 2019 – 2024

B.S. Molecular & Biomedical Biology, Minors: Chemistry & Mathematics

- Spent six weeks and a couple thousand dollars trying to get a simple gene deletion mutant, only to realize I had been using an incompatible conjugation donor strain the entire time.

WORK EXPERIENCE

Graduate Research Assistant *Carja and Bridges Labs, Carnegie Mellon University, 2024 – Present*

- Trained vision transformer for two weeks on images that had been inadvertently normalized twice; model achieved excellent performance predicting preprocessing artifacts.

PhD Rotation Student

Cooper Lab, Univ. of Pittsburgh, 2024

- An undergraduate I was helping accidentally put flaming hot tweezers in 70% ethanol and set an invisible fire on the benchtop; (no comment).

Crew Member & Certified Trainer

Chipotle Mexican Grill, Omaha, NE, 2020 – 2023

- Got distracted by Kylian Mbappe's second goal in the 2022 World Cup final and spilled a bag of sour cream on the floor and myself; no customers were harmed.
- Gave myself mild hand arthritis scooping & mashing thousands of avocados a day.

TECHNOLOGIES I HAVE BROKEN

Programming: Python (works on my machine), Julia (abandoned mid-project after three months of investment), R (switched back to Python), Git (has force-pushed to main; recovered)

Biology: qRT-PCR (makes me want to die), gel electrophoresis (have definitely never run a gel in water instead of buffer), mammalian cell culture (shockingly no issues here), bioinformatics (ran pipeline on wrong reference genome for a week)

COURSE PROJECTS (NULL RESULTS)

Predicting Fitness Effects of Synonymous Mutations *10-710 Computational Genomics, CMU, 2025*

Built an "interpretable" CNN predicting fitness effects of synonymous mutations in *S. cerevisiae*; model was ultimately less interpretable than the raw data. GPA survived.

Probing Chirality in Boltzmann Generators

MSCBIO2030 Comp. Structural Biology, Pitt, 2025

Studied ability of SO(3)-equivariant Boltzmann generators to capture chirality in dipeptide systems. Generators were not cooperative. Poster reframed as investigating "fundamental limitations."

ACTIVITIES I HAVE ATTEMPTED

Bouldering (2024) — tried it for a year, got to V4, broke both my shin bones requiring two surgeries, 17 screws, 2 plates, an external fixator (google it at your own peril) and couldn't walk or stand for 6 months.

Sousaphone player, Omaha Central High School Marching Band (2016–2019) — fell during a performance marching backwards while wearing a 50lb brass sousaphone, causing two other sousaphone players to fall on top of me (not unlike dominos).